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CBA-Whisper: Curriculum Learning-Based AdaLoRA Fine-Tuning on Whisper for Low-Resource Dysarthric Speech Recognition

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Abstract

Whisper is a powerful automatic speech recognition (ASR) model. However, its zero-shot performance on low-resource speech requires further improvement, especially in dysarthric speech recognition (DSR). This paper addresses the Interspeech 2025 Speech Accessibility Project Challenge (SAPC) by fine-tuning Whisper large-v2 on SAP 2024-04-30, UA-Speech, and TORGO datasets using adaptive low-rank adaptation (AdaLoRA). We incorporate improved WhisperX processing and rule-based postprocessing designed for stuttering and machine hallucination. In addition, we employ curriculum learning (CL) with adaptively optimized data filtering to progressively enhance the performance of our model. Using less than 1/5 of the official training data, our final system ranked 2nd in this challenge, with a WER of 10.51% and a SemScore of 85.50% on Test 2 Split, reducing WER by 41.02% and improving SemScore by 12.72% over the baseline (WER 17.82%, SemScore 75.85%). Code and models are publicly available at https://github.com/tan90xx/BCA-whisper.

Index Terms: automatic speech recognition, parameter-efficient fine-tuning, curriculum learning, dysarthric speech

1. Introduction

In recent years, Automatic Speech Recognition (ASR) has achieved unprecedented progress, driven by the advancements in both training data and model architectures, exemplified by systems like OpenAI's Whisper large [1], Google's Universal Speech Model (USM) [2], Facebook's SeamlessM4T [3], and NVIDIA's Canary-1B [4]. Among these ASR models, the Whisper model has gained significant attention as a versatile, off-the-shelf solution, inspiring the development of various adaptations. These include Distil-Whisper [5], WhisperX [6], Whispering-LLaMA [7] and Crisper-Whisper [8], each addressing specific challenges and extending its utility for specific use cases. However, despite the ubiquity of these speech technologies, individuals with speech disabilities often remain excluded from benefiting from these advancements. This disparity stems from the fact that most ASR systems are primarily designed to accommodate mainstream speaking norms, which are typically based on high-resource conditions involving commonly accepted age groups, regions, and cultures. Consequently, low-resource scenarios, such as dysarthric speech and other speech disorders, have been largely unaddressed.

Dysarthric speech, a representative type of low-resource speech, is a motor speech disorder caused by conditions such as Parkinson's disease (PD), multiple sclerosis (MS), and other neurological impairments, which affect the control of speech-related organs [9]. The unique features of dysarthric speech, such as poor articulation, decreased volume and clarity, slower speaking rates and increased dysfluencies, not only pose significant challenges for ASR systems, but also heighten the likelihood of machine hallucinations, a phenomenon where the model generates text inconsistent with the actual input. Recent studies have shown that ASR models, including Whisper, are particularly prone to generating hallucinations when processing speech from individuals who have a lowered ability to express themselves using speech and voice [10]. This is because the model struggles to interpret irregular speech patterns, stuttering repetitions, and prolonged pauses, often filling in gaps with fabricated or irrelevant content. Such hallucinations can produce harmful outputs, such as perpetuating violence, making inaccurate associations, or implying false authority, thereby exacerbating the risks for vulnerable users in real-world applications.

To this end, the performance of dysarthric speech recognition (DSR) remains insufficient for widespread adoption. A key challenge stems from the scarcity of rich and diverse dysarthric speech datasets. Prior to the release of SPA [11], only a limited number of publicly available datasets existed, such as UA-Speech [12], which is restricted to isolated words, and TORGO [13], which comprises data from only 8 dysarthric speakers. These limitations hinder the development of robust models due to the high variability of dysarthric speech, both within and across individuals, which complicates adequate representation in ASR corpora and impairs model generalization and performance. Despite these challenges, recent advancements demonstrate the potential of leveraging SPA datasets to improve DSR. For instance, fine-tuning the wav2vec 2.0 model on the SAP 2023-10-05 has yielded significant improvements, with WER reductions of 37.62% and 26.97% compared to fine-tuning on 100h and 960h of LibriSpeech data, respectively [14]. These results highlight the utility of the SPA dataset in enhancing DSR performance and underscore the importance of high-quality, domain-specific data. However, while the value of data in improving DSR is evident, how to effectively utilize limited data remains an under-explored aspect of research.

Current approaches to improving automatic speech recognition (ASR) often focus on enhancing feature representations through model architecture innovations or large-scale data collection. Examples include spectro-temporal deep features [15] and pre-trained self-supervised learning (SSL) features [14, 16, 17, 18, 19]. While these methods have demonstrated notable improvements, they often fail to fully leverage the state-of-the-art large-scale pre-trained ASR models, such as Whisper. As a result, these approaches tend to hit performance ceilings more quickly, limiting their ability to address complex and low-resource speech recognition tasks effectively.

Regarding Whisper fine-tuning methods, the most common way is to fine-tune all the parameters (full fine-tuning). How-

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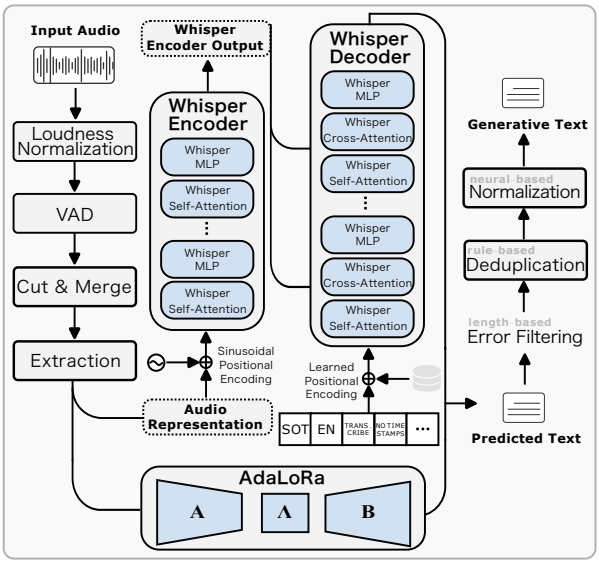


Figure 1: *High-level overview of CBA-Whisper.*

ever, Whisper pre-trained models typically incurs large memory footprint. For example, Whisper turbo consists up to 809 million parameters; Whisper large contains up to 1550 million parameters. As a result, full fine-tuning is typically performed on Whisper small (244 million parameters) to mitigate memory constraints [20, 21]. Recently, parameter-efficient fine-tuning (PEFT) [22] techniques have been applied to the fine-tuning of Whisper models. Prompt tuning [23] and low-rank adaptation (LoRA) [14] fall under this category. LoRA introduces lightweight, task-specific adapters that modify only a small subset of model parameters, allowing the core model to remain unchanged. Building upon LoRA, AdaLoRA [24] further improves efficiency by adaptively allocating rank budgets across different layers, leading to better performance with even fewer updated parameters [25].

Regarding domain adaptation training, strategies for maximizing the utility of low-resource, diverse datasets have received less attention. If information about the hallucinations is known and used for adaptive training, it should enhance the "Careless Whisper" [10]. Curriculum Learning (CL) is well-suited for this scenario, as it systematically organizes training data to facilitate gradual adaptation. CL posits that progressing from learning simple data to more complex data is similar to the human learning process. Hsieh et al. [26] integrate CL by incorporating regrouping and leveraging articulatory, speaker, and speech intelligibility features. However, their approach requires careful data selection, which may limit its scalability and ease of application.

With this perspective in mind, in this paper we design CBA-Whisper based on the Whisper large-v2, fine-tuning it on the subsets of SAP 2024-04-30, UA-Speech, and TORGO datasets using AdaLoRA. We integrate the improved WhisperX preprocessing pipeline and apply rule-based postprocessing specifically designed for stuttering and machine hallucination. Additionally, we employ CL with a more streamlined solution by filtering out data using SemScore [27] after preliminary fine-tuning, ensuring a more practical and effective approach. The main contributions of this paper are summarized as follows:

- Present the CL-based AdaLoRA fine-tuning (CBA) method, which reduces parameters and data required while accelerating training speed, making it ideal for low-budget settings.
- Propose a novel CL strategy for ASR fine-tuning, which optimizes the use of multiple existing publicly available dysarthric speech datasets and mitigates hallucination.
- Introduce a robust preprocessing and postprocessing framework to reduce hallucination and repetition errors.
- Surprisingly, we achieve 2nd place using less than 1/5 of the official training data. This demonstrates that efficient utilization of data, rather than sheer quantity, can lead to outstanding performance.

2. Overview

Fig 1 depicts a high-level overview of CBA-Whisper, which consists of the following three components: (1) Preprocessing, where we apply loudness matching, voice activity detection (VAD), and a cut-and-merge operation to segment audio into approximately 30-second input chunks; (2) Whisper large-v2 with AdaLoRA, where we apply PEFT following the Whisper Finetune Repository[1]; and (3) Postprocessing, where we combine length-based error filtering, rule-based deduplication, and neural-based normalization.

Apart from the SPA 2024-04-30 dataset, we employ UA-Speech and TORGO datasets to supplement our training data. However, after the SAPC concluded, we discovered an issue with data transfer that had gone unnoticed during the challenge. This resulted in the use of only 53.48 hours of the SPA 2024-04-30 Train split dataset instead of the originally planned 290 hours. We provide a detailed description of the datasets used in our experiments below and publicly release the training details, in order to make our results reproducible and facilitate further experimentation in the field.

- **SAP-Dev -** We constructed a subset of the SAP 2024-04-30 Dev set, termed SAP-Dev, comprising the first 1,000 audio samples (totaling 2.34 hours). This dataset was consistently used for evaluation during the training process.
- **SAP-Train -** We used the 53.48-hour subset of the SPA 2024-04-30 Train set and the remaining 41.23-hour subset of the SPA 2024-04-30 Dev set as our SAP-Train dataset. This combined dataset was selected and consistently utilized for training throughout the entire training process.
- **UA-Speech -** The total duration of UA-Speech dysarthric dataset is 48.44 hours. We collected the 6.92-hour subset of UA-Speech for the out-of-domain supplementary data according to the adaptive optimized rule presented in Sec 3.2.
- **TORGO -** The total duration of TORGO dysarthric dataset is 2.02 hours. We collected the 0.13-hour subset of TORGO for the out-of-domain supplementary data according to the adaptive optimized rule presented in Sec 3.2.

2.1. Preprocessing

Loudness Normalization

Individuals with speech disorders, such as dysarthria, often exhibit reduced vocal loudness, which may be attributed to a lack of confidence in producing accurate sounds or to physical limitations that prevent them from sustaining typical speaking volumes. We apply loudness normalization based on the ITU-R BS.1770-4 standards[2] to ensure consistent loudness levels

[1] https://github.com/yeyupiaoling/Whisper-Finetune
[2] https://github.com/fishaudio/audio-preprocess

in dysarthric speech transcription. Our preprocessing ensures that both peak amplitude and perceptual loudness remain within controlled ranges. Both the training and inference datasets undergo loudness matching, improving the robustness of DSR.

VAD Cut & Merge

While the majority of the data durations are concentrated below 5 seconds, all datasets contain samples with durations exceeding 30 seconds (the maximum processing duration of Whisper is 30 seconds). This highlights the necessity of adopting the WhisperX [6] framework, not only for this challenge but also for real-world applications. Long-form audio processing is crucial, especially considering that individuals with dysarthria tend to speak at a slower pace compared to typical speakers. Building upon WhisperX, we incorporate augmentation and adopt a more relaxed VAD threshold from the default{onset: 0.500, offset: 0.363} to {onset: 0.300, offset: 0.300}, allowing for improved detection of speech segments to mitigate cases where transcription outputs are empty.

2.2. Whisper with AdaLoRA Transcription

The resulting speech segments are now approximately equal to the model's input size. During parallel transcription, the decoder generates tokens sequentially based on the modified parameters:

$$h_t = \text{TransformerDecoder}(y_{<t}, \mathbf{X}; \mathbf{W}_0 + \Delta\mathbf{W}) \quad (1)$$

$$\Delta\mathbf{W} = \mathbf{A}\Lambda\mathbf{B} \quad (2)$$

where $y_{<t}$ is the generated text sequence at time step t; $\mathbf{X}$ represents the extracted features of the input speech; $\mathbf{W}_0$ is the pre-trained Whisper weight matrix; $\Delta\mathbf{W}$ is the adaptive low-rank update learned during fine-tuning. $\mathbf{A} \in \mathbb{R}^{d_1 \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d_2}$ are low-rank matrices, whose rank is adaptively allocated across layers. $\Lambda \in \mathbb{R}^{r \times r}$ is a diagonal matrix, contains r singular values, with $r \ll \min\{d_1, d_2\}$. Each element Λ_i corresponds to the scaling factor for the respective low-rank update at layer i.

We perform decoding without conditioning on previous text, using Whisper's no timestamp decoding method. The temperature is restricted to only 2 choices $\{0.0, 0.1\}$ rather than 10, to mitigate hallucination and repetition[3].

2.3. Postprocessing

Error Filtering

This is a simple error detection method, which considers a specific scenario: when the length of a single word exceeds 15 characters, it is highly likely to be a result of machine hallucination. This assumption is based on the general observation that words longer than 15 characters are uncommon in natural language. While this is a rule-based approach, it is particularly effective in addressing machine hallucinations involving the repetitive occurrence of individual letters.

Text Deduplication

The process involves two steps: word-level and phrase-level deduplication. At the word level, if erroneous word is detected, a regular expression is applied to identify and remove repeated substrings. If a repeating unit is found, the word is replaced with its non-repeating component; otherwise, it is discarded. At the phrase level, if the input text contains more than three words, consecutive repeated phrases are detected and reduced to a single instance. This two-step approach ensures the removal of redundant patterns at both the word and phrase levels, resulting in cleaner and more concise text.

[3]Refer to: https://github.com/openai/whisper/discussions/81

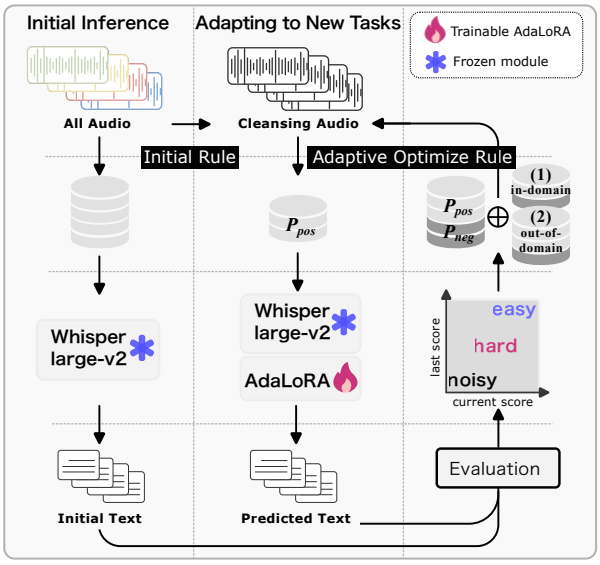


Figure 2: *Schematic diagram of CL-based AdaLoRA fine-tuning (CBA) training strategy.*

Text Normalization

The text normalization is performed using the Normalizer module from the NeMo [28] with the following configurations: the input case is preserved, the language is set to English, and punctuation is post-processed. This ensures standardized and consistent formatting of the text for evaluation.

3. Curriculum Learning Approaches

Our approach consists of three steps designed to optimize model performance: (1) We utilize Whisper large-v2 to generate initial text outputs. (2) Fine-Tuning with Initial Rule: We apply the CBA-Whisper model, trained on a selected dataset using the initial rule, as detailed in the section of Data Cleansing, to produce predicted text for new tasks. (3) Fine-Tuning with Adaptive Optimization Rules: By comparing the generative text from step (1) and (2), we identify discrepancies and refine the training dataset for the next iteration of task adaptation in step (2). Fig 2 depicts a schematic diagram of CBA training strategy.

3.1. Evaluation

There are two evaluation metrics: WER, and SemScore [27]. WER is the normalized string edit distance between the reference transcript and the speech recognizer hypothesis. SemScore is a weighted combination of BERTscore, phonetic distance, and natural language inference probability.

This paper utilizes three evaluation datasets: **Test1**, **Test2**, and **SAP-Dev**. The results on the Test1 (unshared) subset will be publicly displayed on the leaderboard and may be used for hyperparameter tuning. In contrast, the results on the Test2 (unshared) subset will remain confidential until the competition concludes. Both Test1 and Test2 exhibit a distribution of etiologies similar to that of the SAP 2024-04-30 Train and Dev sets. Additionally, SAP-Dev is a subset we independently partitioned from the SAP 2024-04-30 Dev set, which is used for local testing and validation.

3.2. Data cleansing

Initial Rule

It is designed to filter out sentences that contain special characters, such as parentheses, square brackets, or curly braces, ensuring that only pure English sentences are retained for fine-tuning. This rule ensures that the data used for fine-tuning is clean and free from potentially noisy or non-standard text elements. It is straightforward and easy to implement, providing a reliable initial filtering mechanism without introducing unnecessary complexity.

Adaptive Optimization Rule

The adaptive optimization rule introduces a more nuanced approach to data selection based on SemScore. We use a dual-threshold approach to ensure that only high-quality and semantically relevant data is utilized. (1) For in-domain data (SAP-Dev), the rule is relatively lenient: sentences are discarded only if their SemScore remains below 50 both before and after fine-tuning. This ensures that only noisy data is removed while retaining potentially useful in-domain examples. (2) For additional out-of-domain data (UA-Speech and TORGO), the rule is stricter: sentences are retained only if their SemScore exceeds 50 both before and after fine-tuning, and if there is a positive improvement in the score after fine-tuning. By dynamically adjusting the selection criteria based on data domain and quality, this rule optimizes the fine-tuning process for better generalization and accuracy. We adopt 50 as a global threshold, indicating that the effectiveness of the method does not depend on precise threshold tuning.

4. Results

4.1. Experimental Setup

Training Our models are trained on 4 NVIDIA H20 for up to 2 epochs, with a batch size of 6, a maximum duration of 30 seconds, and a minimum duration of 0.5 seconds. Each model takes around 1 hour to fine-tune Whisper large-v2 from scratch. **Hyperparameters** For all our models, we set the same initial rank $r = 12$ and a target rank of $r = 4$. We employ the initial learning rate for the PEFT model at 1e-3, with 50 warmup steps and use AdamW algorithm to optimize our networks.

4.2. Official Results

Our submitted system ranked 2nd on both the official Test1 and Test2 set. Table 1 shows the official ranking of the top scoring teams. Notably, ngohider had the smallest performance gap between the two test sets, while ours exhibited the largest gap.

Table 1: *Results of the SPAC.*

Rank	Team Name	Test 1		Test2		Diff	
		WER	SemScore	WER	SemScore	WER	SemScore
1	slp-lab	5.97	92.53	8.11	88.43	+2.14	-4.10
2	witty	7.98	90.37	10.51	85.50	+2.53	-4.87
3	takagi	8.43	88.89	10.90	84.42	+2.47	-4.47
4	ngohider	10.06	86.34	11.80	83.29	+1.74	-3.05
5	SAGI	10.15	87.78	12.15	84.08	+2.00	-3.80

4.3. Submission Results

Table 2 presents several currently available open-source solutions [1, 5, 6]. From the Test1 results, we observed that single-language English models outperform multilingual models, and larger models tend to yield better results. Since more hallucinations appeared to be present in the large-v3 than in the v2 during fine-tuning, we opted for large-v2 as the base model, as its performance is comparable. In Table 3, we present our ablation results organized by timeline. It can be observed that applying loudness normalization as a preprocessing step to all audio, along with our proposed pre- and post-processing strategies, consistently improves model performance.

Table 2: *Results of Available Models and Systems.*

Model	Para. (M)	VRAM (GB)	Speed (×)	ET (sec)	Test 1 WER	Test 1 SemScore
tiny [1]	39	~ 1	~ 10	7057	33.50	61.77
tiny.en [1]	39	~ 1	~ 10	6564	26.08	68.40
base [1]	74	~ 1	~ 7	8456	27.17	68.16
base.en [1]	74	~ 1	~ 7	7780	20.87	73.68
turbo [1]	809	~ 6	~ 8	9298	22.48	81.48
distil-large-v2 [5]	756	~ 11	~ 6	6495	15.52	79.69
distil-large-v3 [5]	756	~ 11	~ 6	6551	14.52	81.26
x-large-v2 [6]	1550	~ 10	~ 4	7100	12.18	82.50
x-large-v3 [6]	1550	~ 10	~ 4	7170	11.66	84.04

Table 3: *Results of Submissions from the witty Team*

Method	Test 1		SAP-Dev		Submitted at
	WER	SemScore	WER	SemScore	
step0 finetune	10.02	87.38	9.46	90.61	2025-01-13
+deduplication	9.92	87.99	9.36	90.97	2025-01-19
+increase volume	8.75	88.81	9.78	90.84	2025-01-24
>norm loudness	8.69	88.93	9.54	90.75	2025-01-25
all norm loudness	8.48	89.38	7.67	90.86	2025-01-27
step1 finetune	8.13	89.88	6.39	94.12	2025-01-31
+ post processing	8.09	90.17	6.45	94.07	2025-02-01
step2 finetune	7.98	90.37	6.13	94.00	2025-02-02

4.4. Discussion

Limitation We employ an iterative data selection strategy by continuously comparing model performance before and after fine-tuning, ensuring that only data contributing to performance improvement is retained. This approach has led to a gradual enhancement of our model. However, an important trade-off must be considered: while filtering out noisy data prevents potential degradation in model performance, it also results in the loss of valuable textual information. Consequently, the transcribed text corpus derived from speech recognition may become a subset of the dataset's full text corpus, potentially leading to performance degradation due to reduced linguistic diversity.

A possible solution is synthesizing dysarthric speech from text to preserve the full textual distribution while augmenting the training data. However, generating high-quality synthetic dysarthric speech presents another challenge worth further exploration [29, 30, 31], and our experiment with Fish-Speech [32] fine-tuned synthesis yielded unsatisfactory results.

5. Conclusion

In this paper, we described our CBA-Whisper methods for the SAPC, organized as part of Interspeech 2025. We achieved competitive results, ranking 2nd in the challenge. The proposed approach, based on Whisper large-v2 fine-tuned with AdaLoRA, incorporates WhisperX preprocessing, rule-based postprocessing for stuttering and hallucination, and CL with adaptive data filtering to enhance robustness and accuracy. This work presents a generalizable method that can be broadly applied to low-resource speech ASR fine-tuning tasks.

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